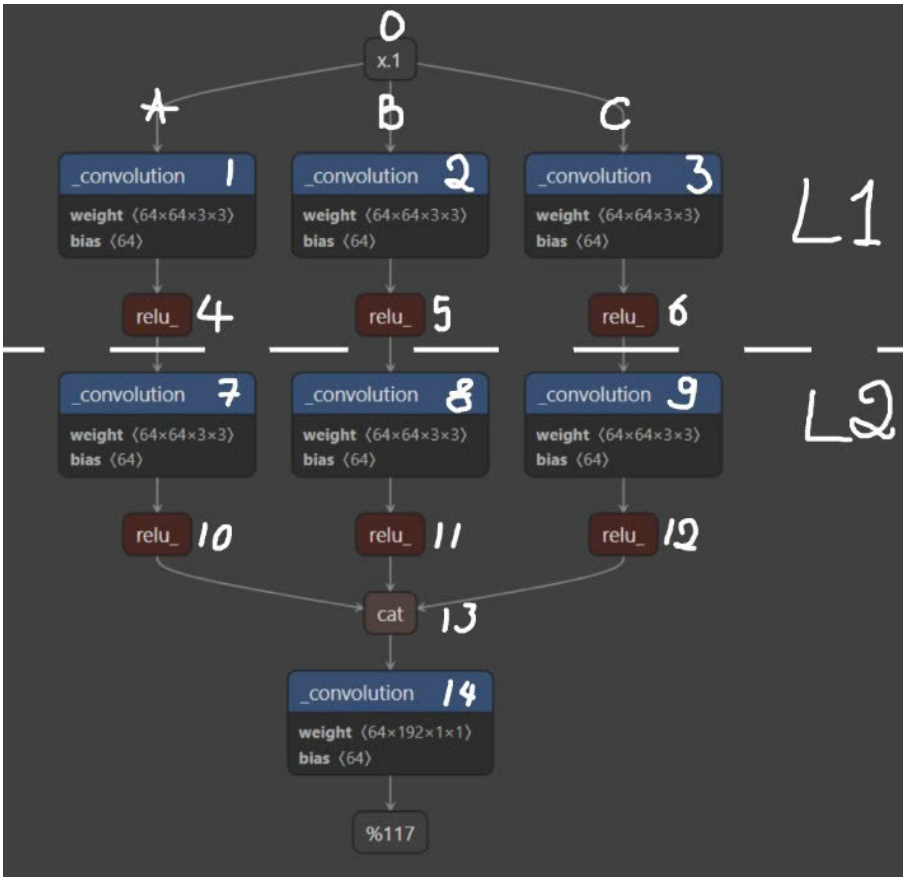
1 DAG Structure



Input: `x.1` — shape $(1, 64, 56, 56)$

All Convolutions are configured with stride $= 1$, dilation $= 1$, kernel $= 3$ (except the final Conv which is 1×1).

2 Stage 1: DAG Construction

2.1 Node Identification

Node ID	Op type	Input tensors	Output tensor	Params
0	`x.1` (input placeholder)	—	$(1, 64, 56, 56)$	—
1	`linalg.conv_2d_nchw_fchw` (Branch A, L1)	`x.1`, `W_A1`	$(1, 64, 56, 56)$	stride $= 1$, dil $= 1$
2	`linalg.conv_2d_nchw_fchw` (Branch B, L1)	`x.1`, `W_B1`	$(1, 64, 56, 56)$	stride $= 1$, dil $= 1$
3	`linalg.conv_2d_nchw_fchw` (Branch C, L1)	`x.1`, `W_C1`	$(1, 64, 56, 56)$	stride $= 1$, dil $= 1$
4	`linalg.elemwise_unary` / ReLU (Branch A)	`T1`	$(1, 64, 56, 56)$	op $=$ relu
5	`linalg.elemwise_unary` / ReLU (Branch B)	`T2`	$(1, 64, 56, 56)$	op $=$ relu
6	`linalg.elemwise_unary` / ReLU (Branch C)	`T3`	$(1, 64, 56, 56)$	op $=$ relu
7	`linalg.conv_2d_nchw_fchw` (Branch A, L2)	`T4`, `W_A2`	$(1, 64, 56, 56)$	stride $= 1$, dil $= 1$
8	`linalg.conv_2d_nchw_fchw` (Branch B, L2)	`T5`, `W_B2`	$(1, 64, 56, 56)$	stride $= 1$, dil $= 1$
9	`linalg.conv_2d_nchw_fchw` (Branch C, L2)	`T6`, `W_C2`	$(1, 64, 56, 56)$	stride $= 1$, dil $= 1$
10	`linalg.elemwise_unary` / ReLU (Branch A)	`T7`	$(1, 64, 56, 56)$	op $=$ relu
11	`linalg.elemwise_unary` / ReLU (Branch B)	`T8`	$(1, 64, 56, 56)$	op $=$ relu
12	`linalg.elemwise_unary` / ReLU (Branch C)	`T9`	$(1, 64, 56, 56)$	op $=$ relu
13	`tensor.concat` / cat	`T10, T11, T12`	$(1, 192, 56, 56)$	dim $= 1$
14	`linalg.conv_2d_nchw_fchw` (1×1 conv)	`T13`, `W_fin`	$(1, 64, 56, 56)$	stride $= 1$, dil $= 1$

2.2 Topological Order (Kahn's Algorithm + Min-Heap)

Topological order: $[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14]$

3 Stage 2: Liveness Analysis

3.1 Tensor Size Calculation

We have Float32 = 4 bytes/element (1 word = 4 bytes).

Tensor Sizes:

- T_{input} (`x.1`): $1 \times 64 \times 56 \times 56 \times 4 = 802,816$ bytes ~ 784 KB
- $T_1 = T_2 = T_3$: $1 \times 64 \times 56 \times 56 \times 4 = 802,816$ bytes ~ 784 KB
- $T_4 = T_5 = T_6$: $1 \times 64 \times 56 \times 56 \times 4 = 802,816$ bytes
- $T_7 = T_8 = T_9$: $1 \times 64 \times 56 \times 56 \times 4 = 802,816$ bytes
- $T_{10} = T_{11} = T_{12}$: $1 \times 64 \times 56 \times 56 \times 4 = 802,816$ bytes
- T_{13} (`cat` output): $1 \times 192 \times 56 \times 56 \times 4 = 2,408,448$ bytes ~ 2.3 MB
- T_{14} (final output): $1 \times 64 \times 56 \times 56 \times 4 = 802,816$ bytes

Weights:

- $W_{Ax}/W_{Bx}/W_{Cx}$ (L1, L2): $64 \times 64 \times 3 \times 3 \times 4 = 147,456$ bytes
- W_{fin} (1×1 conv): $64 \times 192 \times 1 \times 1 \times 4 = 49,152$ bytes

3.2 Computation of $p(t)$ and $\ell(t)$ for Intermediate Tensors

Tensor	Producer (p)	Consumers	$\ell(t) = \max(\text{consumers})$
T_{input}	0	$\{1, 2, 3\}$	3
T_1	1	$\{4\}$	4
T_2	2	$\{5\}$	5
T_3	3	$\{6\}$	6
T_4	4	$\{7\}$	7
T_5	5	$\{8\}$	8
T_6	6	$\{9\}$	9
T_7	7	$\{10\}$	10
T_8	8	$\{11\}$	11
T_9	9	$\{12\}$	12
T_{10}	10	$\{13\}$	13
T_{11}	11	$\{13\}$	13
T_{12}	12	$\{13\}$	13
T_{13}	13	$\{14\}$	14
T_{14}	14	$\{\}$ (sink)	14

3.3 Live Tensors at Each Program Point

Tensor t is alive at program point i if and only if: $p(t) \leq i \leq \ell(t)$

Point i	Live Tensors (Active in Memory)	Total Bytes
0	T_{input}	802,816
1	T_{input}, T_1	1,605,632
2	$T_{\text{input}}, T_1, T_2$	2,408,448
3	$T_{\text{input}}, T_1, T_2, T_3$	3,211,264
4	T_1, T_2, T_3, T_4 ($T_{\text{input}} \to$ dead)	3,211,264
5	T_2, T_3, T_4, T_5 ($T_1 \to$ dead)	3,211,264
6	T_3, T_4, T_5, T_6 ($T_2 \to$ dead)	3,211,264
7	T_4, T_5, T_6, T_7 ($T_3 \to$ dead)	**3,211,264**
8	T_5, T_6, T_7, T_8 ($T_4 \to$ dead)	**3,211,264**
9	T_6, T_7, T_8, T_9 ($T_5 \to$ dead)	**3,211,264**
10	T_7, T_8, T_9, T_{10} ($T_6 \to$ dead)	**3,211,264**
11	T_8, T_9, T_{10}, T_{11} ($T_7 \to$ dead)	**3,211,264**
12	$T_9, T_{10}, T_{11}, T_{12}$ ($T_8 \to$ dead)	**3,211,264**
13	$T_{10}, T_{11}, T_{12}, T_{13}$ ($T_9 \to$ dead)	**4,816,896 (Global Peak)**
14	T_{13}, T_{14} ($T_{10}, T_{11}, T_{12} \to$ dead)	**3,211,264**

*** Notes:**

1. **Points 4, 5, 6:** T_{input} has no consumers after Node 3 ($\ell(T_{\text{input}}) = 3$); hence at $i = 4$, T_{input} is already deallocated. Conversely, T_1 is currently read by Node 4, meaning T_1 remains alive at

$i = 4$. The exact live set at Point 4 is $\{T_1, T_2, T_3, T_4\}$.

2. **Points 7 to 12:** Node 7 reads T_4 to produce T_7; thus at $i = 7$, T_4 is still alive ($p(T_4) \leq 7 \leq \ell(T_4) = 7$). The true live set consists of 4 tensors ($3,211,264$ B), rather than 3 tensors ($2,408,448$ B).

3. **Point 13 (`cat` operator):** Throughout the execution of Op 13, all 3 input tensors (T_{10}, T_{11}, T_{12}) and the output tensor T_{13} must reside simultaneously in memory: $(3 \times 802,816) + 2,408,448 = \mathbf{4,816,896}$ bytes $\approx \mathbf{4.59}$ MB. This represents the true **Global Peak Memory** of the whole model graph.

4. **Point 14 (Conv 1×1):** Op 14 reads T_{13} and generates T_{14}; therefore, memory occupancy is $2,408,448 + 802,816 = \mathbf{3,211,264}$ bytes (not $802,816$ B).

4 Stage 3: Candidate Selection

$$\text{Budget} = 512 \text{ KB} \times 80\% = 409,600 \text{ bytes}$$

Conv L1 (nodes 1, 2, 3)

$$\begin{aligned} \text{op_type} &= \texttt{conv_2d_nchw_fchw} \\ \text{output shape} &: (1, 64, 56, 56) \\ \text{input shape} &: (1, 64, 56, 56) \\ \text{weight shape} &: (64, 64, 3, 3) \end{aligned}$$

Full working set:

$$\begin{aligned} \text{output} &: 1 \times 64 \times 56 \times 56 \times 4 = 802,816 \text{ bytes} \\ \text{input} &: 1 \times 64 \times 56 \times 56 \times 4 = 802,816 \text{ bytes} \\ \text{weight} &: 64 \times 64 \times 3 \times 3 \times 4 = 147,456 \text{ bytes} \\ \hline \text{Total} &: 1,753,088 \text{ bytes} \sim 1.67 \text{ MB} \end{aligned}$$

$1,753,088 > \text{Budget} \implies$ **Tile (Candidate)**

ReLU (nodes 4–6, 10–12)

$$\text{Full working set} = \text{output} + \text{input} = 2 \times 802,816 = 1,605,632 \text{ bytes}$$

$1,605,632 > \text{Budget} \implies$ **Tile (Candidate)**

Conv L2 (nodes 7–9)

$$\text{Total} = 1,753,088 \text{ bytes} > \text{Budget} \implies \textbf{Tile (Candidate)}$$

Cat (node 13) `tensor.concat` is a data movement operation, not a compute `linalg` operator (`is_tileable` exclusion). $\implies$ **Do Not Tile**

Final Conv (node 14)

$$\begin{aligned} \text{output} &: 1 \times 64 \times 56 \times 56 \times 4 = 802,816 \text{ bytes} \\ \text{input} &: 1 \times 192 \times 56 \times 56 \times 4 = 2,408,448 \text{ bytes} \\ \text{weight} &: 64 \times 192 \times 1 \times 1 \times 4 = 49,152 \text{ bytes} \\ \hline \text{Total} &: 3,260,416 \text{ bytes} \sim 3.11 \text{ MB} \end{aligned}$$

$3,260,416 > \text{Budget} \implies$ **Tile (Candidate)**

Final Candidate List

Node	Op	Full WS (bytes)	Tile?
$1, 2, 3$	Conv 3×3 L1	$1,753,088$	✓
$4, 5, 6$	ReLU L1	$1,605,632$	✓
$7, 8, 9$	Conv 3×3 L2	$1,753,088$	✓
$10, 11, 12$	ReLU L2	$1,605,632$	✓
13	cat	N/A	×
14	Conv 1×1	$3,260,416$	✓

5 Stage 4: Dimension Selection

Under EdgeSlice priority order for NCHW layout (dense convolution, batch=1):

$$\text{Spatial H} \succ \text{Spatial W} \succ \text{Channel}$$

$\implies$ **Prioritize slicing along the H dimension (`spatial_h`) for both Conv and ReLU.**

6 Stage 5: Tile Size Calculation

$$\text{Budget} = 409,600 \text{ bytes}$$

6.1 Conv L1, L2

$\text{shape} = (1, 64, 56, 56), \text{dim} = \texttt{spatial_h}, \text{stride} = 1, \text{dilation} = 1, \text{kernel} = 3$

Output tile:

$$(1, 64, T_h, 56) = 64 \times T_h \times 56 \times 4 = 14,336 \times T_h \text{ bytes}$$

Calculating required input height with spatial halo padding:

$$\begin{aligned} h_{\text{in}} &= (T_h - 1) \times \text{stride} + \text{dilation} \times (\text{kernel}_h - 1) + 1 \\ &= (T_h - 1) \times 1 + 1 \times 2 + 1 \\ &= T_h + 2 \quad \text{(Add 2 halo rows)} \end{aligned}$$

Input tile:

$$(1, 64, T_h + 2, 56) = 64 \times (T_h + 2) \times 56 \times 4 = 14,336 \times (T_h + 2) \text{ bytes}$$

Weight tile:

$$(64, 64, 3, 3) \times 4 = 147,456 \text{ bytes}$$

Total footprint:

$$\begin{aligned} F(T_h) &= 14,336 \times T_h + 14,336 \times (T_h + 2) + 147,456 \\ &= 28,672 \times T_h + 28,672 + 147,456 \\ &= 28,672 \times T_h + 176,128 \text{ bytes} \end{aligned}$$

Enforcing capacity constraint $F(T_h) \leq 409,600$:

$$28,672 \times T_h \leq 233,472 \implies T_h \leq 8.143$$

$\implies$ Optimal integer tile size is $T_h = 8$:

$$F(8) = 28,672 \times 8 + 176,128 = 229,376 + 176,128 = 405,504 \text{ bytes} \leq \text{Budget}$$

With $T_h = 8$:

- Each tile produces 8 output rows and requires 10 input rows ($8 + 2$ halo).
- Number of tiles $= \lceil 56/8 \rceil =$ **7** tiles.

6.2 All ReLU Layers

Analogously (no halo expansion and no weight parameters):

$$F(T_h) = 28,672 \times T_h$$

Constraint: $F(T_h) \leq 409,600 \implies T_h \leq 14.28 \implies$ best: $T_h = 14 \xrightarrow{\text{align to power-of-2}} \mathbf{T_h = 8}$

- Number of tiles $= \lceil 56/8 \rceil =$ **7** tiles.

6.3 Final Conv (Node 14, Conv 1×1)

Analogously (1×1 pointwise conv requires no spatial halo):

$$\begin{aligned} \text{Input tile} &: (1, 192, T_h, 56) = 192 \times T_h \times 56 \times 4 = 43,008 \times T_h \text{ bytes} \\ \text{Output tile} &: (1, 64, T_h, 56) = 64 \times T_h \times 56 \times 4 = 14,336 \times T_h \text{ bytes} \\ \text{Weight tile} &: (64, 192, 1, 1) \times 4 = 49,152 \text{ bytes} \\ \hline F(T_h) &= (43,008 + 14,336) \times T_h + 49,152 = 57,344 \times T_h + 49,152 \text{ bytes} \end{aligned}$$

Constraint: $57,344 \times T_h + 49,152 \leq 409,600 \implies 57,344 \times T_h \leq 360,448 \implies T_h \leq 6.285$.

$$\text{best: } T_h = 6 \xrightarrow{\text{align to power-of-2}} \mathbf{T_h = 4}$$

Tile size for Final Conv: $\mathbf{T_h = 4}$
Number of tiles $= \lceil 56/4 \rceil =$ **14** tiles

7 Stage 6: Schedule Search

Given 1D cascade search and 1 legal dimension per op:

Node	Op	Slice dim	Tile size	Tiles
$1, 2, 3$	Conv 3×3 L1	H	8	7
$4, 5, 6$	ReLU L1	H	8	7
$7, 8, 9$	Conv 3×3 L2	H	8	7
$10, 11, 12$	ReLU L2	H	8	7
14	Conv 1×1	H	4	14

8 Stage 7: Fusion Planning

8.1 Fusion Chains

Final anchors:

- $\{1, 2, 3\}$ (Conv L1 per branch)
- $\{7, 8, 9\}$ (Conv L2 per branch; ReLU L1 is fused into here)
- $\{10, 11, 12\}$ (ReLU L2)
- $\{14\}$ (Conv 1×1)

8.2 Re-fit Tile Size after Fusion

Group members: Node 7 (Conv L2) + Node 4 (ReLU_A1)
Group total footprint:

$$\begin{aligned} F_{\text{group}}(T_h) &= F_{\text{Conv L2}}(T_h) + F_{\text{ReLU}}(T_h) \\ &= (28,672 \times T_h + 176,128) + 28,672 \times T_h \\ &= 57,344 \times T_h + 176,128 \text{ bytes} \end{aligned}$$

Constraint: $57,344 \times T_h + 176,128 \leq 409,600 \implies 57,344 \times T_h \leq 233,472 \implies T_h \leq 4.071$.

$$\text{best: } T_h = 4 \xrightarrow{\text{align}} \mathbf{T_h = 4}$$

Tile size after fusion:

Node	Op	Slice dim	Tile size	Tiles
$1, 2, 3$	Conv 3×3 L1	H	8	7
$7, 8, 9$	Conv 3×3 L2 fused	H	4	14
$10, 11, 12$	ReLU L2	H	8	7
14	Conv 1×1	H	4	14

9 Stage 8: MLIR Transform IR Generation

This transformation script is automatically synthesized by the EdgeSlice compiler. Manual calculation is impractical due to SSA handle chaining.

10 Memory Tiers

Op	**Tile Working Set (Memory Tier)**
Conv L1 (tile $H = 8$)	$405,504$ B $\approx$ 396 KB
Conv L2 fused (tile $H = 4$)	$405,504$ B $\approx$ 396 KB
ReLU L2 (tile $H = 8$)	$14,336 \times 8 \times 2 = 229,376$ B $\approx$ 224 KB
Conv 1×1 (tile $H = 4$)	$\mathbf{278,528}$ B $\approx$ **272** KB $^{(*)}$